

Landline Call Blocker 3 Video Outline

- Uses caller ID to block incoming landline telephone calls
- A telephone line interface circuit detects ring, decodes caller ID, disconnected house phone from line, takes line off hook to implement “answering” the call.
- Caller ID decoder circuit is a Holtek 9032C (datasheet suggests a circuit diagram)
- Caller data stored in EEPROM
- Caller name
- Caller phone number
- Last call Date
- Total call count from this name and number
- Record is a partial duplicate to another record (name OR Number matches an existing record)

Calls are processed based on rules and whether callers already exist in EEPROM memory as a specific call type.

Incoming Call Types

- Allowed: caller name and or number matches a record in EEPROM (whitelist)
- “Voicemail”: first time call from name and or number, house phones are disconnected from line and all may go to service provider voicemail. (type U)
- Blocked call. Name and/or number exist in database as Blocked type B. Relay takes line “off hook” for 4 seconds and then returns to “on hook” (hanging up)
- “State” type spam, name is in the format “CITY ST” where ST is a two digit state code preceded by two blank characters.

Duplicate call records: Incoming caller name & number is not a record in eeprom but name or number matches an existing EEPROM record. This record is marked as “DUP = P”. Otherwise, all records are marked as DUP=0.

Memory Full: When EEPROM memory is full the code will look for the oldest type “U” record and insert the new caller record.

Call Year: Caller ID does not provide the year that the call was made. The code looks for a transition from month 12 to 1 in incoming calls and advances the two digit year value store in processor EEPROM.

Status LED (red, green, yellow)

Steady red = error

Brief Green= incoming called allowed

Brief Yellow= house disconnected (voicemail)

Brief Red= Call blocked (off hook / on hook)

LDC Display

- 4x20 LCD displays caller info and EEPROM call records
- Use “right arrow” to edit call type in eeprom.
- Use “Save “ button to save the edit.
- Backlight an edit mode timeout if unattended

Serial Port

- The code sends call progress messages to the serial port.
- Utility commands are provided and shown below.

Serial Commands

Command	Example	Description
BNK	BNK	Blinks the status LEDs (red & green = yellow)
DAT		Shows the stored year and month.
YER	YER,26	Stores the two-digit year. Your startup log says it is currently unset (255).
DIS	DIS,0	Displays one record # 0.
LST	LST	Lists all non-empty records.
WHT,name,number	WHT,JOHN SMITH,2155551234	Adds a number to the whitelist, or changes its existing record to whitelist. Name: up to 15 characters; number: exactly 10 digits.
ENM,rec num,name	ENM,0,TEST CALLER	Changes a record's caller name.
TYP,record,type	TYP,30,B	Changes a record's type: W whitelist, B blocked, U unknown, or E empty (clears the record)
FMT	FMT	Formats the external EEPROM to empty records
IMP	IMP	Do not use! Un-needed data import feature.

DIS Command output format

Field	Meaning
DUP:0	No partial match was found when that caller was first stored.
DUP:P	A partial match was found: either the name matched another record but the number differed, or the number matched but the name differed. The call was still stored as a separate record.
COUNT:00	Starting count for the imported CB2 records. The CB2 dump did not include their old counts.
COUNT:01 through 99	Number of subsequent calls that exactly matched that stored name and number. It increases on each exact match and stops at 99.

All code generated by ChatGPT based on experience and guidance by author.